

AEGIS: Structural Sensitivity for Adversarial Vulnerability Analysis of GNNs

Anonymous Author(s)

Abstract—Graph neural networks deployed in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, yet no prior method unifies, from one closed-form computation, the three diagnostics a practitioner needs: a globally maximally sensitive perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node first-order sensitivity radii (first-order diagnostics, complementary to sound certificates such as AGNNCert). AEGIS derives all three from one object, the *constrained sensitivity matrix* S_c , computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to $N=7,650$ on a single GPU. The analytical construction applies to any GNN with continuous edge-weight-modulated message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT[†]); predictive transfer is architecture-dependent, with deeper-than-2-layer models transferring most reliably (29/33 architecture-dataset cells positive, sign-test $p < 10^{-5}$). For the spectral-norm-constrained IGNN subclass ($\sim 6\%$ Cora accuracy cost) we additionally derive a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ and a worst-case three-regime characterisation, empirically anchored by a κ_{max} sweep (section V-D). Tightness lies in $[1.00, 1.39]$ over $\varepsilon \in [0.01, 0.20]$ across 7 architectures and 9 datasets; the SVD direction reaches $0.72\text{--}0.92\times$ of an IFT-gradient PGD baseline; non-adaptive top- v_{ij} edge masking reduces SVD damage by $42 \pm 8\%$ vs. $11 \pm 6\%$ random ($p < 0.002$). On IEEE power-flow benchmarks (proof-of-concept, not operator-grade), S_c correlates with N-1 contingency rankings on case14–118 ($\tau = +0.37$ to $+0.62$, $P@10 = 0.66\text{--}0.81$); code will be released under tiered access.

Index Terms—Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial graph [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: which edges, if perturbed, would cause predictions to fail, and by how much?

Structural attacks [1], [4], [5] give per-edge surrogate gradients but no continuous direction or per-node budgets; smoothing [6], [7], [8], [9] and IBP-style certifiers [10], [11] give per-node certificates but no edge rankings or direction; robust-architecture defenses [12], [13], [14] harden models without surfacing a vulnerability map; sober-look critiques [15], [16] flag evaluation pitfalls. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) unifies the three outputs at the level of one analytical object, S_c (section IV); the formal regime guarantees of theorem 1 apply to contractive

implicit GNNs, and S_c extends as a computational tool to any GNN with continuous edge-weight-modulated message passing (section V-F).

Contributions. (1) S_c and a matrix-free pipeline yielding the SVD-optimal direction, per-edge rankings, and per-node radii from one computation, scaling to $N=7,650$ (matrix-free/dense σ_1 agree to 0.03% at $N=200$). (2) A worst-case three-regime characterisation with critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ for the spectral-norm-constrained IGNN subclass ($\sim 6\%$ accuracy cost; theorem 1). (3) Empirical evaluation over 9 datasets, 7 architectures, 4 domains (330 runs), with four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5]/AGNNCert/LODF (table IV), continuous-to-discrete transfer (fig. 7), and a power-grid case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N \times N}$, node features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S \in \mathbb{R}^{Nd \times N^2}$, constrained $S_c \in \mathbb{R}^{Nd \times |E|}$, unrolled $S_K \in \mathbb{R}^{Nd \times N^2}$ (theorem 7), and per-node block S_v ; the contractivity $\kappa = \|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v .

IGNN and IFT. A DEQ-GNN [17], [18] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [19] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized [20] so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point [21], [22] with optional Jacobian regularization [23], and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}, \quad (2)$$

where $\|(I - J_z)^{-1}\|_2 \leq 1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ [24] measures the gap to the spectral-radius bound (with $\eta=1$ for normal J_z). The IFT has been used for hyperparameter optimization [25] and implicit differentiation [21], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \|\delta\hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A} = \delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation attacks; insertion attacks (e.g., Nettek [1], Mettack [4]) require enlarging the basis from E to a candidate set $\bar{E} \supseteq E$ and lie outside our threat model. Discrete deletions are connected to the continuous model through theorem 6. Unlike input-feature perturbation [26], [27], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, which is what makes structural sensitivity qualitatively different and motivates the construction of S_c in section III.

III. ADVERSARIAL EQUILIBRIUM THEORY

Building on the threat model of section II, we now derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which the three diagnostics follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [28] (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14\text{--}0.59$ across datasets; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\text{crit}}$ in our experiments so practitioners can audit. If (A3) fails, AEGIS still returns S_c rankings and SVD-optimal attacks but flags that the formal guarantees below are void.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ exhibit three regimes:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): For worst-case directions that maximally increase $\|\hat{A} + \delta\hat{A}\|_2$ (aligned with the top singular vector of $\hat{A}$), $\|(I - J'_z)^{-1}\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ from the Neumann lower bound $1/(1 - \|J'_z\|_2)$, which holds without

normality assumptions on J'_z . Generic directions diverge more slowly because $\|J'_z\|_2$ need not approach 1.

(c) **Supercritical** ($\varepsilon \geq \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J'_z\|_2$ may exceed 1). The perturbed operator may still be contractive along low-sensitivity directions, but Banach no longer guarantees uniqueness and the part-(a) first-order guarantees lapse; we do not claim divergence as a generic post-threshold behavior, only that our certificate fails (empirical κ_{max} sweep: section V-D).

Proof. (a). Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\|\delta\hat{A}\|$ (the boundary set $\{\hat{A} : z^*(\hat{A}) \text{ lies on a region face}\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [28] yields a conservative IFT on each region. With $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta\hat{A}) + O(\|\delta\hat{A}\|^2), \quad (6)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). Along the worst-case direction $\|J'_z\|_2 \rightarrow \|\hat{A}\|_2 \|W\|_2 + \varepsilon \|W\|_2$, so $1 - \|J'_z\|_2 \geq \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$; the Neumann lower bound $\|(I - J'_z)^{-1}\|_2 \geq 1/(1 - \|J'_z\|_2)$ requires no normality assumption and gives the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ rate. For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach fails; the iteration may converge elsewhere, oscillate, or diverge. $\square$

$\varepsilon_{\text{crit}}$ is sufficient, not necessary, and the unconstrained $\sqrt{r}$ slack is 7–14 $\times$ on 50-node subgraphs; S_c closes this in practice (tightness ≈ 1.00 at $\varepsilon=0.01$; section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [29]; the gap is the pseudospectral index η , bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$: if all activations are positive, $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$. For general ReLU patterns $\eta \leq \kappa(V_{J_z})$, which empirically stays near $\kappa(V_W)$ ($\eta=1.19\text{--}2.47$; section V-E).*

Proof. When $\phi \equiv 1$, $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [29]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}})=1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$ yields $\eta \leq \kappa(V_W)$. Graph structure does not amplify nonnormality; the moderate empirical η traces to the spectral-norm regularisation of W .

Proposition 3 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$,*

where v_1 is the leading right singular vector of S . The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.

By the variational characterisation of σ_1 , the direction is optimal for hidden-state damage and empirically aligned with classification damage (section V-B). The constrained sensitivity matrix $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top) / \sqrt{2}$ ($\|b_k\|_F = 1$), so $\sigma_1(S_c)$ is consistent with $\|\delta \hat{A}\|_F = \varepsilon$. The $N^2 \rightarrow |E|$ projection P_c is the standard duplication-matrix reduction [30] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The envelope ratio is tight (≈ 1.00) at $\varepsilon=0.01$ and reaches ≤ 1.39 at $\varepsilon=0.20$, identifying the dangerous direction rather than a global safety margin.

Proposition 4 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c \neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta \hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v . Under the constrained threat model the tighter radius $r_v^{(c)} = m_v / (\|W_{y_v} - W_{c^}\|_2 \|S_{c,v}\|_2) \geq r_v$ holds, using the block-rows of S_c at node v .*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \cdot \text{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|W_{y_v} - W_{c^*}\|_2 \|S_v\|_2 \|\delta A\|_F$; misclassification requires $|\Delta f_{y_v}| \geq m_v$, and inverting gives r_v . Intuitively, r_v is the margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structural perturbation); high-margin, low-sensitivity nodes get large radii.

Remark 5 (Sensitivity threshold semantics). r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v , and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [6] certifies which nodes are guaranteed safe, while AEGIS differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon=0.01$ and rises with ε as expected for first-order approximations (section V-B).

A. Continuous-to-Discrete Transfer

Proposition 6 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \| [S_c]_{:,k} \|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from full removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \quad (8)$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\text{vec}(A)$.

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (9)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1 - \kappa)^2 / L_J$.

Proof. (a). Edge removal corresponds to $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. For the remainder, let $g(t) = z^*(A + t\delta \hat{A}_k)$, $t \in [0, 1]$. IGNN is piecewise affine in A for a fixed activation pattern, so $g(t)$ is piecewise linear and crosses finitely many activation boundaries; on each segment J_z is constant. Across a boundary J_z jumps by at most $\|W\|_2^2$ (the max change in $\text{diag}(\phi')(\hat{A} \otimes W)$ when one activation flips), so the worst-case path Lipschitz constant $L_J \leq \|W\|_2^2$. The resolvent identity $\frac{d}{dt}(I - M)^{-1} = (I - M)^{-1} \frac{dM}{dt} (I - M)^{-1}$ then yields $\|g''(\xi)\| \leq (1 - \kappa)^{-2} L_J w_k^2$, hence $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1 - \kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. $\square$

Empirically $|R_k|$ is $2\text{--}10\times$ smaller than $L_J w_k^2$, and the 29/33 positive-transfer cells of fig. 7 confirm the piecewise structure does not erode the bridge: shallow models with near-uniform v_k (GCN-2) violate part-(b), deeper models satisfy it. The ingredients of theorem 1 are classical [29], [24]; the contribution is the constrained projection S_c that turns a vacuous bound (tightness 0.31) into a tight one (1.00; section V-A) and unifies attacks, rankings, and radii.

B. Generalization to Explicit GNNs

Observation 7 (Structural Sensitivity for K -Layer GNNs). *Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix*

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) *The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (11)$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 4) apply identically with S_K in place of S .*

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs inherit

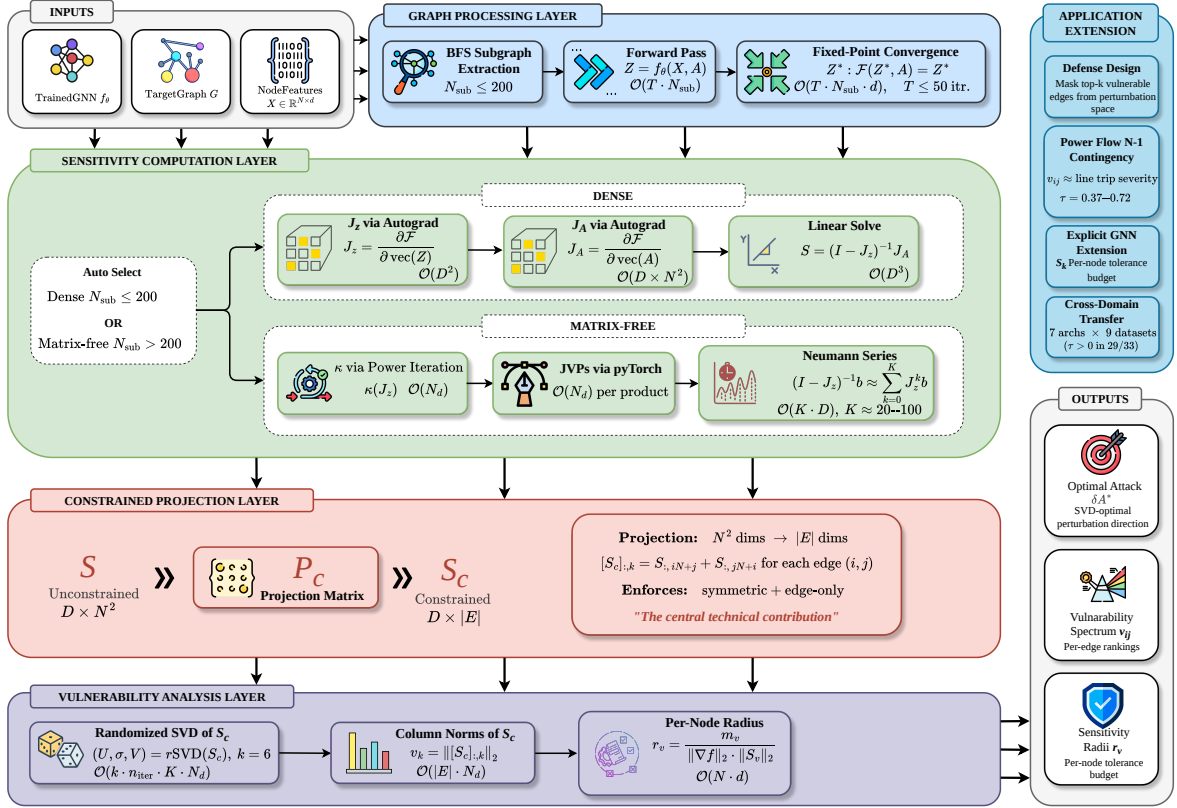


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJP without materialising it, scaling to $N=7,650$.

the practical tools but lack $\varepsilon_{\text{crit}}$ and the regime characterisation, which depend on (A3); a Jacobian product $\prod_k \|J_z^{(k)}\|_2 > 10$ diagnoses unreliable first-order predictions, and the empirical tightness 0.99–1.02 (section V-F) validates the approximation regardless of the a priori bound.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd \times N^2}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta \hat{A}^*$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^K J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6} \|b\|$); each Neumann term and $J_A \text{vec}(\delta A)$ are forward-mode JVPs in $\mathcal{O}(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A

and solves via `torch.linalg.solve`. The rSVD [31] ($k=p=10, n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) for $\delta \hat{A}^* = \varepsilon \cdot \text{sym}(P_c v_1)$; column norms give $\{v_{ij}\}$, and per-node radii follow theorem 4 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39-0.50$; fig. 2) is why one closed-form SVD matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $\mathcal{O}(K \cdot Nd)$ time and $\mathcal{O}(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $\mathcal{O}((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (PyTorch [32]). IGNN [19] uses $F(Z) = \text{ReLU}(\hat{A} Z W^T + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [20] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and 2-head readout. Subgraphs: 50-node BFS from the highest-degree node. The spectral-norm constraint costs IGNN $\sim 6\%$ accuracy (77.5% vs. $\sim 83\%$ on Cora), the price of the guarantees in theorem 1. Datasets: Cora, Citeseer, Pubmed [33], Amazon Photo [34], WikiCS [35], IEEE 14/30/57/118 [36], [37]. Baselines: smoothing (Gaussian $\sigma=0.01$, 200 MC, Clopper–Pearson $\alpha=0.001$), Mettack with Meta-Self GCN surrogate, LODF as $\text{PTDF}_{ij}/(1 - \text{PTDF}_{jj})$

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \leq 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: $F, z^*, \hat{A}$, budget ε ; SVD rank k , oversampling p , power iters n_{iter} ; Neumann tol. τ ; edge set E ; (opt.) head h , logits Y , labels y .

Ensure: $\sigma_1(S_c), \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}}$.

```

1:  $\kappa \leftarrow \|J_z\|_2$  via power iteration;  $K \leftarrow \lceil \log(1/\tau) / \log(1/\kappa) \rceil$ .
2:  $\text{MATVEC}(v) := \sum_{j=0}^K J_z^j \text{JVP}_{\hat{F}}^{\hat{A}}(P_c v)|_{z^*}$ ;
    $\text{RMATVEC}(u) := P_c^\top \text{VJP}_{\hat{F}}^{\hat{A}}(\sum_{j=0}^K (J_z^\top)^j u)|_{z^*}$ .
3: Draw  $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$ ;  $Y \leftarrow \text{MATVEC}(\Omega)$  column-wise.
4: for  $j = 1, \dots, n_{\text{iter}}$  do
5:    $Y \leftarrow \text{QR}(\text{MATVEC}(\text{RMATVEC}(Y)))$ 
6: end for
7:  $Q \leftarrow \text{QR}(Y)$ ;  $B^\top \leftarrow \text{RMATVEC}(Q)$ ;  $(\tilde{U}, \Sigma, V^\top) \leftarrow \text{SVD}(B)$ ;  $U \leftarrow Q\tilde{U}$  {rSVD [31]}
8:  $\sigma_1 \leftarrow \Sigma_{1,1}$ ;  $\delta \hat{A}^* \leftarrow \varepsilon \cdot \text{sym}(P_c V_{:,1}) / \|\text{sym}(P_c V_{:,1})\|_2$ .
9:  $v_{ij} \leftarrow \|\text{MATVEC}(e_{ij})\|_2$  for each  $(i, j) \in E$  {per-edge spectrum}
10: if  $h, Y, y$  supplied then
11:    $r_v \leftarrow m_v / (\|W_{y_v} - W_{c^*}\|_2 \|S_{c,v}\|_2)$  for each  $v$  via 10 power iters {Prop. 4}
12: end if
13: if  $\|W\|_2$  extractable then  $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$  {Thm. 1}
14: return  $(\sigma_1, \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$ .
```

from the DC B -matrix [38, Ch. 11]. Code and checkpoints will be released under tiered access (attack-generation behind institutional-affiliation review).

A. Cross-Domain Adversarial Analysis

Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnerability*; on Cora ($N=2,708$), 50-node rankings correlate weakly with full-graph rankings ($\tau=0.16$; section V-E), so graph-level assessment uses the matrix-free pipeline (section V-D). All bounds use the operator-norm $\kappa = \|J_z\|_2$; on our suite, the κ -based $\varepsilon_{\text{crit}}$ runs up to 53% tighter than the spectral-radius counterpart (worst on Citeseer; pseudospectral index $\eta=1.19$ –2.47). Table I verifies (A3) on all benchmarks ($\kappa < 1$) and reports the per-dataset critical budget $\varepsilon_{\text{crit}}$ and attack advantage.

Table II reports tightness degradation across the five datasets and four ε values.

The envelope ratio is tight (1.00–1.02) at $\varepsilon=0.01$ and grows to 1.05–1.39 at $\varepsilon=0.20$; the bound is reliable only at small ε and under-predicts by up to 39% at the largest budget. The ratio is measured on equilibrium shift $\|\Delta z^*\|_F$, distinct from the prediction-flip (breach) rate of fig. 4 (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at $\varepsilon=0.20$).

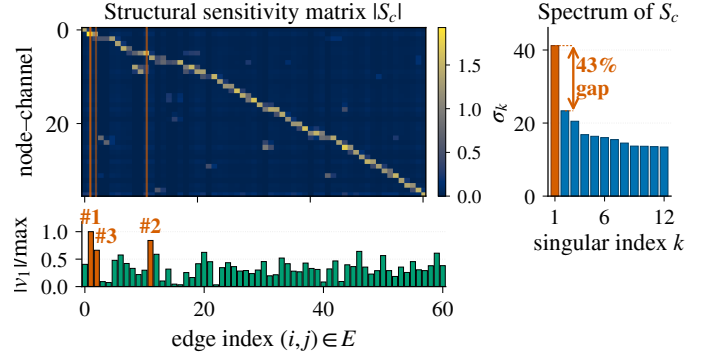


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50$, $d=16$, $|E|=61$, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, $\sigma_1=41.2$, 43% gap to σ_2 (cross-benchmark $(\sigma_1 - \sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes with positive first-order radius r_v .

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	81 $\pm$ 9
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	83 $\pm$ 4
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	74 $\pm$ 23
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	92 $\pm$ 4
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	70 $\pm$ 3

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–10 $\times$ more damage than Mettack [4] (149/150 wins, one-sided sign-test $p<10^{-43}$); GR-BCD/PR-BCD [5] are in table IV. The distinctive value appears under discrete removal (fig. 3).

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Against randomized smoothing [6] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds), base smoothing radii are constant per node ($r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$), e.g. 0.123 vs. AEGIS’s 0.046 at $\sigma=0.05$) and offer no per-edge differentiation; localized smoothing [8] gives per-node deterministic-radius probabilistic certificates at 10^3 – 10^4 MC samples per node but still lacks an edge structure or attack direction. AEGIS radii are structurally informative ($r_v \approx 0.09$ in dense regions, 0.01 at boundaries), placing the three methods at complementary points on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). Three observations follow: 1,000 random unit directions reach only 0.45–0.49 $\times$ of $\sigma_1(S_c)$, confirming the SVD direction is well separated; an equilibrium-shift PGD attacker [39] with AEGIS’s IFT gradients (50 steps)

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c -optimal constrained perturbation). Tight at $\varepsilon=0.01$ (≈ 1.00); under-predicts by up to 39% at $\varepsilon=0.20$.

Dataset	$\varepsilon=.01$	$\varepsilon=.05$	$\varepsilon=.10$	$\varepsilon=.20$
Cora	$1.01 \pm .00$	$1.07 \pm .01$	$1.15 \pm .03$	$1.36 \pm .08$
Citeseer	$1.01 \pm .00$	$1.07 \pm .01$	$1.16 \pm .03$	$1.39 \pm .07$
Pubmed	$1.01 \pm .00$	$1.07 \pm .01$	$1.15 \pm .02$	$1.38 \pm .06$
WikiCS	$1.00 \pm .01$	$1.01 \pm .01$	$1.02 \pm .01$	$1.05 \pm .01$
Amazon	$1.00 \pm .00$	$1.01 \pm .00$	$1.03 \pm .01$	$1.07 \pm .01$

TABLE III: Equilibrium damage at $\varepsilon=0.10$ across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70 \pm .79$	$2.51 \pm .48$	$3.05 \pm .70$	$0.97 \pm .25$
Citeseer	$4.63 \pm .71$	$2.97 \pm .42$	$3.35 \pm .57$	$1.01 \pm .15$
WikiCS	$2.10 \pm .22$	$0.67 \pm .11$	$1.93 \pm .20$	$0.53 \pm .09$

recovers only 72–92% of the closed-form damage; and a classification-loss PGD through the unrolled iteration inflicts 15–70% less equilibrium damage while flipping predictions comparably (0–1.8%), so the SVD direction flips predictions effectively even when optimising a proxy. We therefore describe the SVD direction as the *maximally sensitive perturbation direction* rather than “the optimal attack”.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. See fig. 3 for four sequential orderings.

Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$); recomputing S_c iteratively after each removal closes ≤ 3 pp of the static-vs-greedy gap on Cora and ≤ 1 pp on Citeseer/Pubmed at $k \times$ compute, so we keep the static ranking as the headline.

Breach rates. Figure 4 reports prediction-flip fractions across ε . Citeseer and WikiCS stay below 2% across the full range; Cora reaches 7.6% at $\varepsilon=0.20$; Pubmed is the right-skewed outlier ($27.4 \pm 21.1\%$ mean, median 7.8% vs. mean 10.3% at $\varepsilon=0.10$). Every breached node satisfies $\varepsilon > r_v$, consistent with r_v as a first-order diagnostic rather than a probabilistic certificate (theorem 5).

C. Comparison with Prior Structural Baselines

Table IV reports head-to-head numbers against GR-BCD [5] (PR-BCD is the same family’s larger-budget variant), AGNNCert [10], and LODF [38]. AEGIS is a label-free closed-form proxy; the question is how much of each gold-standard signal it retains.

TABLE IV: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text); AGNNCert [10] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [38] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD [5]	Cora $k=5$	0.643	1.207	+0.16
AGNNCert [†] [10]	Cora med. r_v	0.187	1.414	+0.08
LODF [38]	case118 P@10	0.81	≤ 0.20	+0.14

[†]AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 5). The numerical gap reflects different mathematical objects, not framework dominance.

GR-BCD [5] converges with S_c on Pubmed ($\tau=+0.69$) and diverges on Cora ($\tau=+0.16$), the dataset-level τ acting as a self-diagnostic; PR-BCD [5] (same family, OGB-scale budget) sits beyond our matrix-free boundary. AGNNCert [10]’s IBP radii are 4.9–10.2 $\times$ looser numerically but mathematically stronger (sound certificate vs. first-order threshold). LODF [38] retargeted to thermal-overload peaks at P@10=0.60 on case57 and collapses on case118; Ejebe–Wollenberg PI [40] is positively correlated but loses to AEGIS on both grids.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds each): the theoretical $\varepsilon_{\text{crit}}$ drops 230 $\times$ but $\rho(J_z)$ saturates at ≈ 0.42 (the ReLU pattern blocks the Jacobian from its theoretical maximum), so the resolvent rises only 1.17 $\rightarrow$ 1.80 instead of the $\sim 70 \times$ that $1/(1 - 0.99)$ predicts. The phase transition is therefore a theoretical safety boundary, not an empirically observable discontinuity; training converges throughout the sweep, though the post-hoc AEGIS analysis exceeds the 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$, per-edge $\tau=0.999$, and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite. For larger N the rSVD error [31] is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$, degrades to 1.031 at $N=200$, coverage stays 82–89%. We use $N=50$ as default (66 $\times$ faster than $N=200$, tightness 1.013 ± 0.003). Subgraph-vs-full-graph agreement is high on dense grids (case14 $\tau=0.80$ –0.86 at $N_{\text{sub}}=8$ –12; case30 $\tau=0.78 \pm 0.07$) but low on citation-scale graphs (Cora $\tau=0.16 \pm 0.13$ since a 50-node BFS covers $\sim 1.8\%$ of edges), so citation-scale users should prefer the full-graph pipeline. BFS-centre choice barely matters ($\tau=0.95 \pm 0.02$ across highest-vs-median-degree). On the full citation graphs (matrix-free), the AtkAdv ratio over degree-proportional *grows* to 3.25 $\times$ (Cora) and 9.82 $\times$ (Cite-

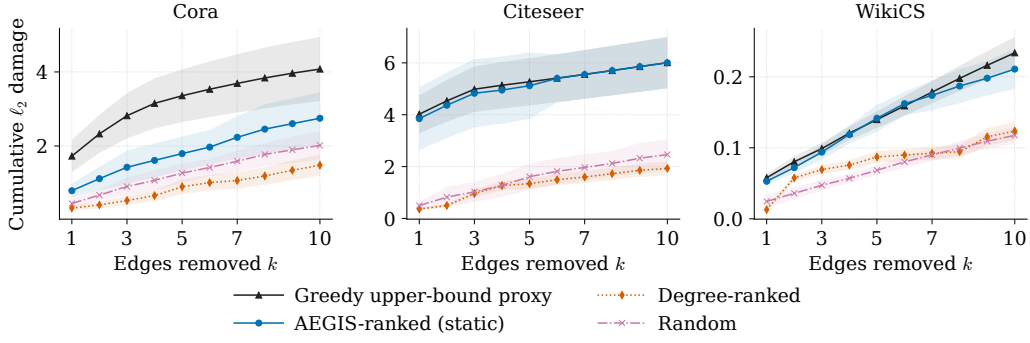


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS matches Greedy on Citeseer/WikiCS and reaches 54–67% of Greedy on Cora at $k=5$ –10; degree-ranked stays below random on Cora at every k .

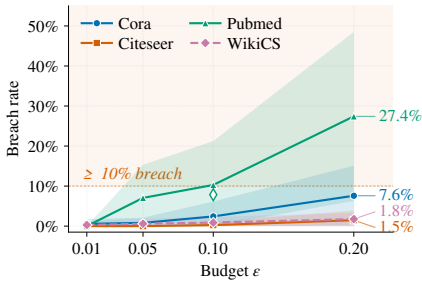


Fig. 4: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ε (10 seeds, mean ± 1 sd bands).

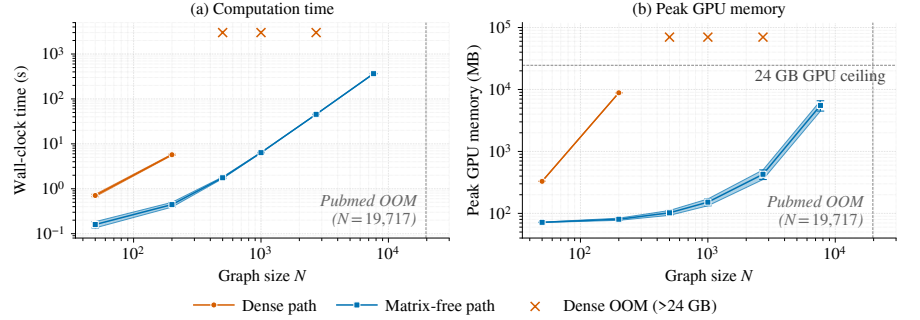


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s/5.5 GB.

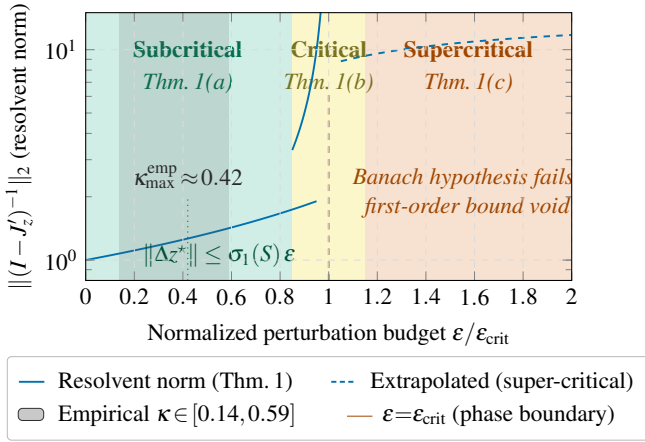


Fig. 6: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1 - \kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

seer) at $k=10$ vs. $\approx 1.1\times$ on 50-node subgraphs, so the BFS regime is conservative rather than artefactual.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$

traces the accuracy–robustness frontier ($r_v=0.090@72.1\%$ vs. $0.051@80.6\%$).

Defense-informed edge protection (non-adaptive). Masking the top- k edges by v_{ij} (Cora IGNN, $N=50$, 10 seeds) cuts SVD damage by $42 \pm 8\%$ at $k=5$ and $61 \pm 7\%$ at $k=10$, vs. $11 \pm 6\%/18 \pm 9\%$ for random masking (paired Wilcoxon $p < 0.002$); the attacker is held fixed, so an adaptive variant recomputing S_c would partially erode the gain.

F. Extension to Explicit GNNs

For K -layer explicit GNNs we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and construct S_c from S_K identically (theorem 7). Table V reports the per-architecture characterisation on Cora: IGNN carries theorem 1; the explicit models receive the computational tool without the regime characterisation.

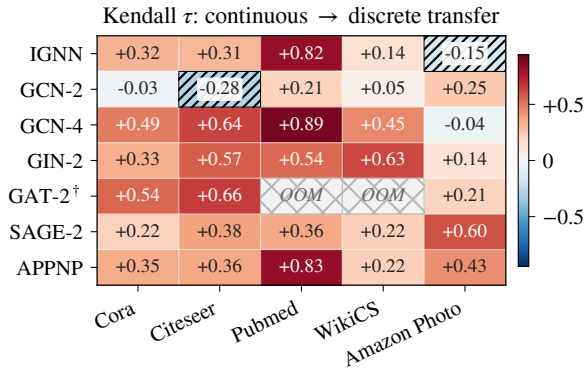
GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined. Our GAT[†] variant modulates attention by the edge weight,

$$h'_i = \sum_{j \in \mathcal{N}(i)} \hat{A}_{ij} \cdot \alpha_{ij} \cdot W h_j, \quad (12)$$

restoring continuous differentiability with respect to $\hat{A}_{ij}$ and yielding tightness 0.99, AtkAdv $4.4\times$, $\tau = +0.36$. Architectures using binary adjacency masks (hard attention, max/min

TABLE V: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall vs. brute-force single-edge-removal truth. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+ .32 $\pm$.04	77.5 $\pm$ 1.7%
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	−.04 $\pm$.03	78.9 $\pm$ 0.9%
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+ .49 $\pm$.02	78.3 $\pm$ 1.8%
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+ .33 $\pm$.03	76.6 $\pm$ 1.9%
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+ .54 $\pm$.06	77.8 $\pm$ 1.2%
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+ .22 $\pm$.10	77.5 $\pm$ 1.5%
APPNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+ .35 $\pm$.01	82.2 $\pm$ 0.6%



[†] GAT-2: 2-layer attention; OOM on Pubmed and WikiCS.

Fig. 7: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (10 seeds, 330 runs). Cell values are mean Kendall τ ; buckets: strong ($\tau > 0.5$), moderate (0.3–0.5), weak (0–0.3), anti-correlated ($\tau < 0$, diagonal hatch); cross-hatch: GPU OOM. Deeper models (GCN-4, APPNP) and GAT[†] dominate.

aggregation) and GATv2 [41]’s dynamic-attention masking fall outside the framework; the architecture-agnostic property applies only to continuous edge-weight-modulated message passing.

Robust backbones. Under the same $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite [12] attains $\tau = +0.37/+0.54$ (Cora/Citeseer) and GNNGuard-lite [13] $+0.10/+0.53$, matching or exceeding IGNN; the cap is a precondition—without it $\kappa > 1$ voids theorem 1.

Cross-dataset transfer. Figure 7 reports τ over 5 datasets and 7 architectures (330 runs); 29/33 cells positive (one-sided sign test $p < 10^{-5}$). Cold cells (GCN-2 on sparse citation graphs, IGNN on dense product graphs at 50 nodes) recover on full-graph matrix-free analysis ($N=7,650$ shifts τ from -0.14 to $+0.24$). Heterophilic WebKB [42] GCN-4 attains $\tau = +0.21/+0.44$.

Theorem 6’s sufficient condition holds for 47–62% of edge pairs; GCN-2 fails the bridge because 2-hop aggregates produce near-uniform first-order shifts. Gradient-based explainers [43] against the same truth give $\tau = -0.07$ to -0.21 ; the theorem 7 bound is $1.4\text{--}5.9\times$ loose, tighter for shallower models.

VI. CASE STUDY: POWER FLOW CONTINGENCY

As a proof-of-concept cross-domain demonstration (not an operator-grade screening tool), we apply S_c to AC power flow. The correspondence is empirical, not a physical derivation: PTDF/LODF [38] are functions of the DC susceptance matrix B , while v_{ij} is derived from the GNN’s learned IFT-resolvent $(I - J_z)^{-1}$. The structural form mirrors DC line-outage factors, but the GNN’s J_A is not B^{-1} ; the observed correlation reflects the fidelity of the GNN’s learned physics. Recent GNN-PF work [44], [45], [46], [3] optimises a fidelity objective; we take the vulnerability-analysis view.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W , trained on five IEEE test cases [36] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower’s [37] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P , Q , $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. For case14–118 we use full-graph dense analysis; for case300 a 200-node BFS subgraph. LODF is benchmarked off-spec on this AC voltage-angle truth (its native metric is DC line-flow change), so the comparison is informative on rank order but disadvantages LODF by construction.

table VI reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain (≈ 1.00), and on the credible range (case14–118) ranking correlation is $\tau = +0.37$ to $+0.62$ with $P@10 = 0.66\text{--}0.81$, identifying 7–8 of the top-10 critical lines. Case300 stress-tests the GNN’s learning capacity, not S_c scalability: at θ RMSE 0.394 p.u. ($\approx 22.6^\circ$) the surrogate has not learned AC physics; its apparent $\tau = +0.72$, $P@10 = 0.87$ are conditioned on an unconverged model and excluded from headline numbers. The matrix-free S_c pipeline itself scales to $N=7,650$ (fig. 5); improved PF surrogates would extend the demonstration into the operational range.

Baselines and stability. LODF (industry DC screening) attains $\tau = 0.44\text{--}0.58$ on credible cases, behind AEGIS’s $0.62\text{--}0.67$ on case57/118 (Wilcoxon $p < 0.01$); CIs overlap on case14/30. The runtime trade-off is unfavorable for operations at this scale (LODF < 0.13 s; brute-force N-1 $0.1\text{--}2$ s; AEGIS $2\text{--}23$ s with training), and operators have impedance data, so the demonstration’s value is conceptual: a label-free GNN proxy recovers the dominant N-1 severity pattern from one closed-form SVD pass over a learned representation. v_1 captures one direction, so it surfaces multi-edge vulnerabilities only partially (top-5 overlap 40–64% with N-2 critical sets edge-wise, 7–18% pair-wise). Pairwise rank stability is $+0.40\text{--}+0.78$, higher on larger grids. *Binary adjacency beats admittance-weighted* ($P@10 = 0.81$ vs. 0.27 on case118; Spearman between admittance magnitude and N-1 severity is near zero, -0.01 to $+0.15$): a single-line trip removes the line regardless of impedance, so binary sensitivity correctly models the all-or-

TABLE VI: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ : Kendall correlation against brute-force N-1; P@10: top-10 precision. Case14–118 is the credible reported range; case300 is a scalability stress-test (model un-converged, not an operational point).

Case	N	$ E $	Model (p.u.)		AEGIS	
			$ V $	θ	τ	P@10
<i>Credible operating range</i>						
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$
<i>Scalability stress-test (model un-converged)</i>						
case300 [‡]	300	411	.031	.394[*]	$+.72_{\pm.03}$	$.87_{\pm.06}$

[‡]200-node BFS subgraph; 1,000 training samples.

* θ RMSE $\approx 22.6^\circ$: not a valid operational data point.

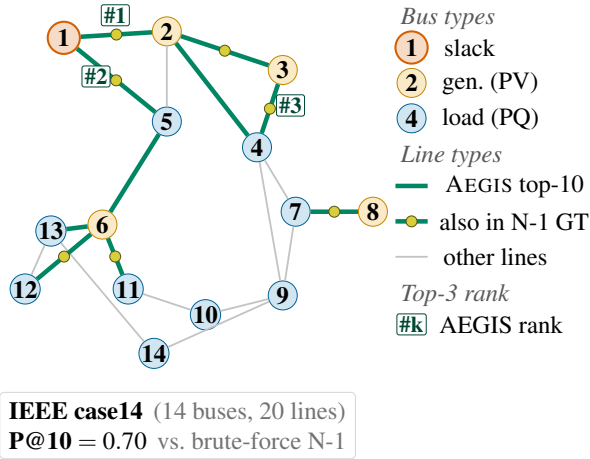


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; P@10 = 0.70 for this run, consistent with the 0.74 ± 0.12 in table VI). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one closed-form SVD pass.

nothing character of N-1. This does not mean impedance is unimportant; sophisticated edge-feature GNNs with impedance, thermal rating, and voltage level as edge attributes are the right next step. Operator-grade deployment requires the extensions noted in the conclusion.

VII. RELATED WORK

We position AEGIS against four threads; each supplies at most one of the three outputs we unify under a single matrix-free pass.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5], and variants [47] optimise surrogate gradients to flip predictions but provide no closed-form globally maximally sensitive direction. Sober-look studies [15], [16] flag evaluation pitfalls; the defence-informed-masking ablation (section V-E) uses paired Wilcoxon against random masking for this reason.

Certified robustness and defense. Convex relaxations [11], smoothing [6], [7], [8], collective certificates [9], and IBP [10] certify per-node robustness but do not identify the vulnerability-driving edges. Empirical defenses [12], [14], [13] harden models without surfacing a vulnerability map. r_v is a first-order threshold, not a sound certificate (theorem 5); the two are complementary.

Implicit networks and influence functions. DEQs [17], IGNN [19], and well-posedness analyses [26] address input sensitivity $\partial z^*/\partial x$; AEGIS targets structural sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ (implicit; theorem 1) and the unrolled Jacobian S_K (explicit; theorem 7). The closed-form derivative descends from influence-function implicit differentiation [25], extended to adjacency under a constrained projection; GNN explainers [43] target prediction fidelity rather than adversarial vulnerability, and we draw on classical matrix perturbation theory [29], [24].

VIII. CONCLUSION

AEGIS unifies per-edge vulnerability rankings, the SVD-optimal attack direction, and per-node first-order sensitivity radii under one constrained sensitivity matrix S_c , computed matrix-free for any GNN with continuous edge-weight-modulated message passing and supplemented by regime guarantees (theorem 1) for contractive implicit models.

Limitations. (i) Radii r_v are first-order sensitivity thresholds, not probabilistic certificates (theorem 5). (ii) Standard GAT and binary-adjacency-mask architectures fall outside the framework; the seventh architecture is the edge-weighted GAT[†] variant (section V-F). (iii) On large citation networks 50-node subgraph rankings correlate weakly with full-graph rankings ($\tau=0.16$ on Cora); the matrix-free pipeline is the recommended graph-level path. (iv) Spectral-norm-constrained IGNN (77.5% Cora) lags explicit GNNs ($\sim 82\%$); the formal track is a model-class trade-off, not a universal cost. (v) Power-flow $\tau=+0.37$ to $+0.62$ is proof-of-concept; operator-grade deployment requires admittance-weighted edges, broader operating coverage, and thermal/voltage-violation severity. (vi) The threat model is edge-deletion / weight-perturbation; insertion attacks (Nettack-style) need a candidate-edge basis $\bar{E}$ extension. The vulnerability map informs defence design (section V-E); code will be released with a 90-day stakeholder notification and tiered-access for attack-generation.

REFERENCES

- [1] D. Zügner, A. Akbarnejad, and S. Günnemann, “Adversarial attacks on neural networks for graph data,” in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2847–2856.
- [2] H. Dai, H. Li, T. Tian, X. Huang, L. Wang, J. Zhu, and L. Song, “Adversarial attack on graph structured data,” in *International Conference on Machine Learning*, 2018, pp. 1115–1124.

- [3] A. M. Nakiganda, C. Weeraddana, N. Popli, and T. van der Hagen, "Graph neural networks for fast contingency analysis of power systems," *arXiv preprint arXiv:2310.04213*, 2023.
- [4] D. Zügner and S. Günnemann, "Adversarial attacks on graph neural networks via meta learning," in *International Conference on Learning Representations*, 2019.
- [5] S. Geisler, T. Schmidt, H. Sirin, D. Zügner, A. Bojchevski, and S. Günnemann, "Robustness of graph neural networks at scale," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 7637–7649.
- [6] A. Bojchevski, J. Klicpera, and S. Günnemann, "Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more," in *International Conference on Machine Learning*, 2020, pp. 1003–1013.
- [7] A. Bojchevski and S. Günnemann, "Certifiable robustness to graph perturbations," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [8] J. Schuchardt, Y. Scholten, A. Bojchevski, and S. Günnemann, "Localized randomized smoothing for GNNs," in *International Conference on Machine Learning*, 2023.
- [9] J. Schuchardt, A. Bojchevski, J. Klicpera, and S. Günnemann, "Collective robustness certificates: Exploiting interdependence in graph neural networks," in *International Conference on Learning Representations*, 2021.
- [10] Y. Li *et al.*, "AGNNcert: Deterministic certification for graph neural networks against adversarial perturbations," in *International Conference on Learning Representations*, 2025.
- [11] D. Zügner and S. Günnemann, "Certifiable robustness and robust training for graph convolutional networks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 246–256.
- [12] D. Zhu, Z. Zhang, P. Cui, and W. Zhu, "Robust graph convolutional networks against adversarial attacks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 1399–1407.
- [13] X. Zhang and M. Zitnik, "GNNGuard: Defending graph neural networks against adversarial attacks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9263–9275.
- [14] W. Jin, Y. Ma, X. Liu, X. Tang, S. Wang, and J. Tang, "Graph structure learning for robust graph neural networks," in *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2020, pp. 66–74.
- [15] F. Mujkanovic, S. Geisler, S. Günnemann, and A. Bojchevski, "Are defenses for graph neural networks robust?" in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 8940–8952.
- [16] L. Gosch, S. Geisler, D. Sturm, and S. Günnemann, "Adversarial training for graph neural networks: Pitfalls, solutions, and new directions," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [17] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [18] S. Bai, V. Koltun, and J. Z. Kolter, "Multiscale deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5238–5250.
- [19] F. Gu, H. Chang, W. Zhu, S. Sojoudi, and L. El Ghaoui, "Implicit graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 11 984–11 995.
- [20] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *International Conference on Learning Representations*, 2018.
- [21] S. Gould, R. Hartley, and D. Campbell, "Deep declarative networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3988–4004, 2021.
- [22] S. W. Fung, H. Heaton, Q. Li, D. McKenzie, S. Osher, and W. Yin, "JFB: Jacobian-free backpropagation for implicit networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 6, 2022, pp. 6648–6656.
- [23] S. Bai, V. Koltun, and J. Z. Kolter, "Stabilizing equilibrium models by Jacobian regularization," in *International Conference on Machine Learning*, 2021, pp. 554–565.
- [24] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.
- [25] J. Lorraine, P. Vicol, and D. Duvenaud, "Optimizing millions of hyperparameters by implicit differentiation," in *International Conference on Artificial Intelligence and Statistics*, 2020, pp. 1540–1552.
- [26] L. El Ghaoui, F. Gu, B. Travacca, A. Askari, and A. Tsai, "Implicit deep learning," *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 3, pp. 930–958, 2021.
- [27] M. Revay, R. Wang, and I. R. Manchester, "Lipschitz bounded equilibrium networks," in *arXiv preprint arXiv:2010.01732*, 2020.
- [28] J. Bolte and E. Pauwels, "Conservative set valued fields, automatic differentiation, stochastic gradient descent and deep learning," *Mathematical Programming*, vol. 188, pp. 19–51, 2021.
- [29] G. W. Stewart and J.-g. Sun, *Matrix Perturbation Theory*. Academic Press, 1990.
- [30] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, 3rd ed. Wiley, 2019.
- [31] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.
- [32] A. Paszke, S. Gross, F. Massa, A. Lerner, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [33] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Galligher, and T. Eliassi-Rad, "Collective classification in network data," *AI Magazine*, vol. 29, no. 3, pp. 93–106, 2008.
- [34] O. Shchur, M. Mumme, A. Bojchevski, and S. Günnemann, "Pitfalls of graph neural network evaluation," in *Relational Representation Learning Workshop, NeurIPS*, 2018.
- [35] P. Mernyei and M. Caron, "Wiki-CS: A Wikipedia-based benchmark for graph neural networks," in *arXiv preprint arXiv:2007.02901*, 2020.
- [36] IEEE PES, "IEEE PES test feeder working group: Power system test cases," *IEEE Power and Energy Society*, 2024, <https://electricgrids.engr.tamu.edu>.
- [37] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dolichon, F. Meier, S. Meinecke, and M. Braun, "pandapower – an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.
- [38] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. John Wiley & Sons, 2014.
- [39] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018.
- [40] G. C. Ejebe and B. F. Wollenberg, "Automatic contingency selection," *IEEE Transactions on Power Apparatus and Systems*, no. 1, pp. 97–109, 1979.
- [41] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *International Conference on Learning Representations*, 2022.
- [42] H. Pei, B. Wei, B. Chang, Y. Lei, and B. Yang, "Geom-GCN: Geometric graph convolutional networks," in *International Conference on Learning Representations*, 2020.
- [43] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [44] B. Donon, B. Donnot, I. Guyon, and A. Marot, "Graph neural solver for power systems," in *International Joint Conference on Neural Networks*, 2019, pp. 1–8.
- [45] V. Bolz, J. Rueß, and A. Zell, "Power flow approximation based on graph convolutional networks," in *18th IEEE International Conference on Machine Learning and Applications*, 2019, pp. 1679–1686.
- [46] C. Kim, T. Conrad, R. Karim, J. Oelhaf, D. Riebesel, T. Arias-Vergara, A. K. Maier, J. Jäger, and S. Bayer, "Physics-informed GNN for medium-high voltage AC power flow with edge-aware attention and line search correction operator," *arXiv preprint arXiv:2509.22458*, 2025.
- [47] H. Wu, C. Wang, Y. Tyshetskiy, A. Docherty, K. Lu, and L. Zhu, "Adversarial examples on graph data: Deep insights into attack and defense," in *International Joint Conference on Artificial Intelligence*, 2019, pp. 4816–4823.